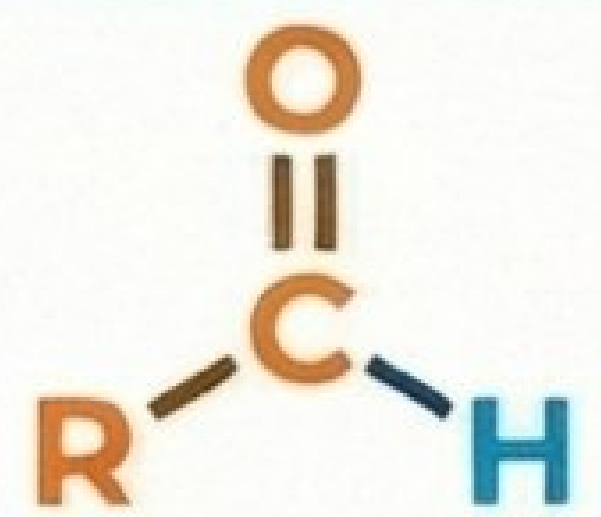
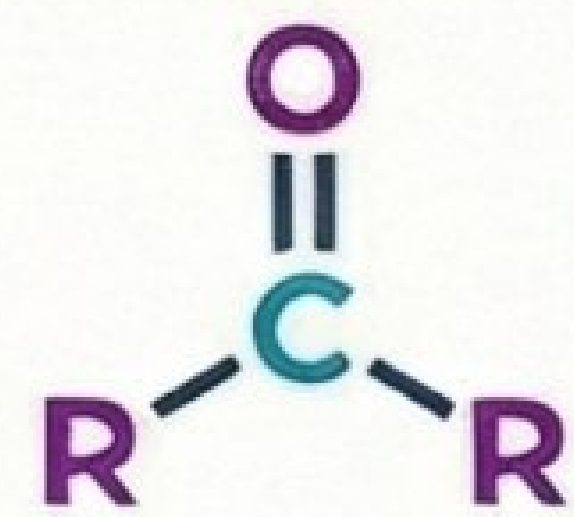


A Visual Guide to Aldehydes, Ketones & Carboxylic Acids

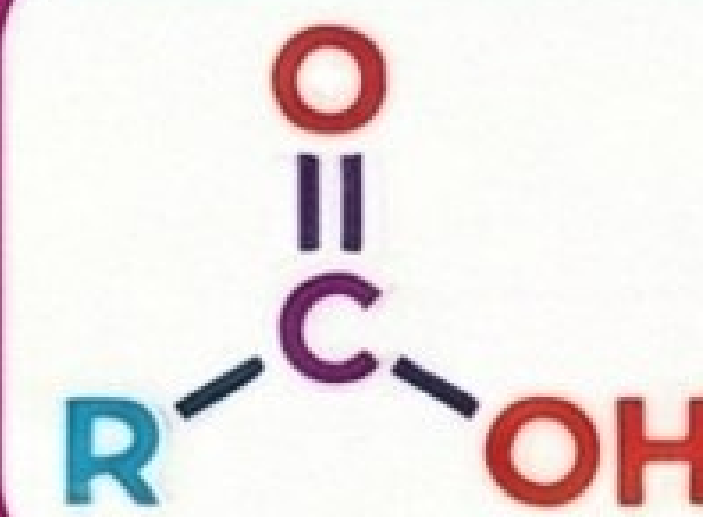


ALDEHYDE:
The "-al" Family
Carbonyl (C=O) bonded to at least one H. Suffix "-al".
e.g., Methanal, Ethanal.

**IDENTIFYING
THE KEY
PLAYERS**



KETONE:
The "-one" Family
Carbonyl (C=O) bonded to two carbon atoms. Suffix "-one".
e.g., Propanone/Acetone.



CARBOXYLIC ACID:
The "-oic acid" Family
Carboxyl (C=O) attached to hydroxyl (-OH), Suffix "-oic acid".
e.g., Ethanoic Acid.

PREPARATION OF ALDEHYDES

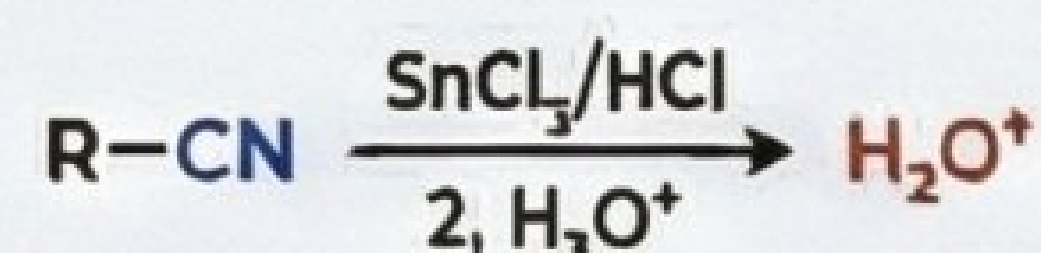


Reisenmund Reaction



Acyl Chloride hydrogenated using H_2 with Pd/BaSO₄ catalyst.

Stephen Reaction

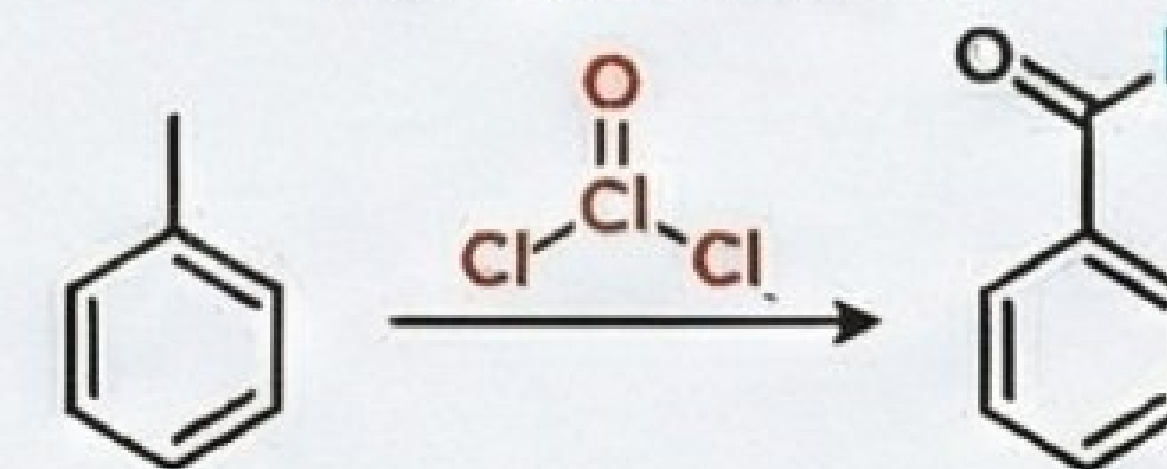


Nitriles reduced with SnCl₄/HCl then hydrolyzed. Mnemonic: "Stephen is a Russian (R CN) Scientist".

Alternative to Stephen Reaction

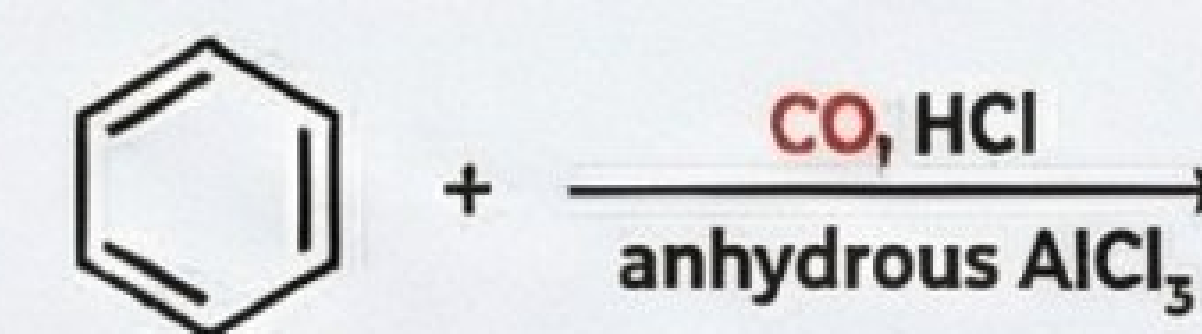
DIBAL H followed by hydrolysis can also convert nitriles to aldehydes.

Etard Reaction



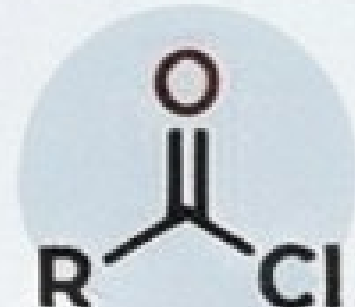
Toluene is oxidized by chromyl chloride (CrO₂Cl₂) to Benzaldehyde.

Cattermann-Koch Reaction



Benzene treated with CO, HCl in presence of AlCl₃.
Mnemonic: A drain cuser (Geffer) looks like Benzene, containing black CO gas and acidic HCl.

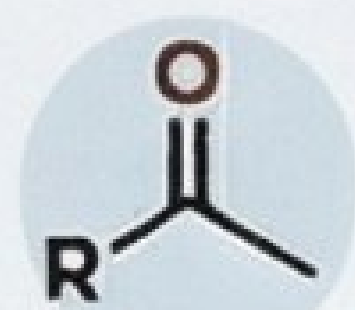
PREPARATION OF KETONES



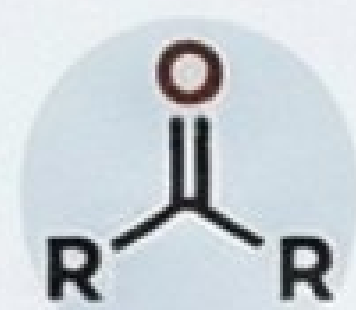
From Acyl Chlorides
Two moles of Acyl Chloride react with Dialkyl Cadmium (R₂Cd) to form a ketone.



From Nitriles
Nitriles react with a Grignard Reagent (R-MgX) followed by hydrolysis (H₂O) to yield a ketone.



From Nitriles
Nitriles react with a Grignard Reagent (R-MgX) followed by hydrolysis (H₂O) to yield a ketone.

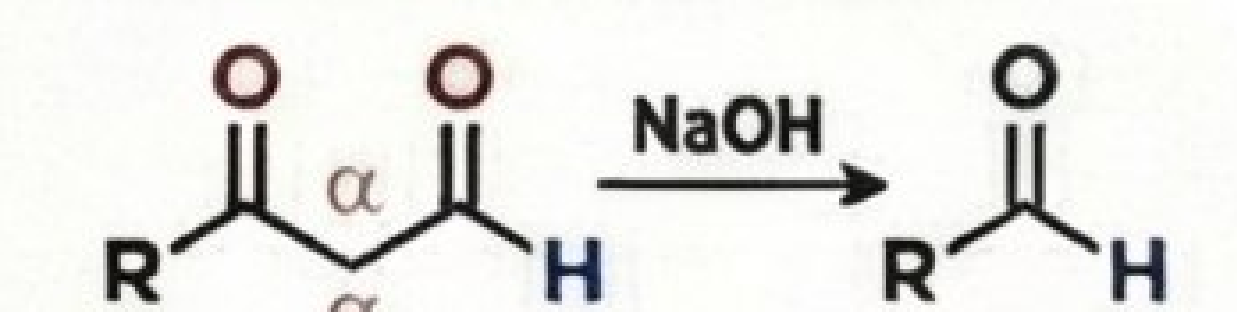


From Nitriles
Nitriles react with a Grignard Reagent (R-MgX) followed by hydrolysis (H₂O) to yield a ketone.

KEY REACTIONS & TESTS

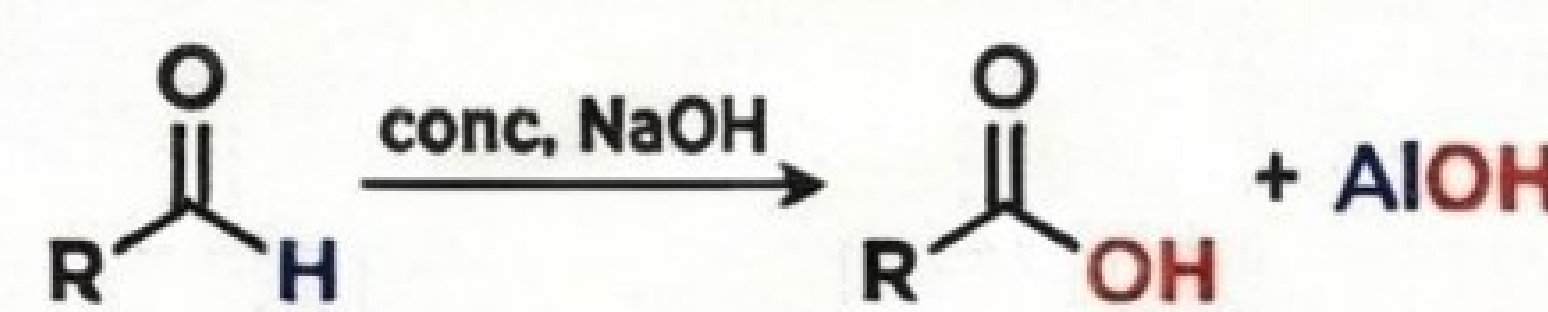
	Tollens' Test	Fehling's Test	Iodoform Test
Test	Ammoniacal Silver Nitrate [Ag(NH ₃) ₂] ⁺ OH ⁻	Fehling's Solution (Cu ²⁺ ions)	I ₂ and NaOH
Outcome	Aldehydes Only 	Aliphatic Aldehydes Only 	Compounds with CH ₃ -C=O group
Fact	Both aliphatic & aromatic aldehydes give this test.	Aromatic aldehydes (like Benzaldehyde) do NOT give this test.	Used to detect methyl ketones and acetaldehyde.

Aldol Condensation



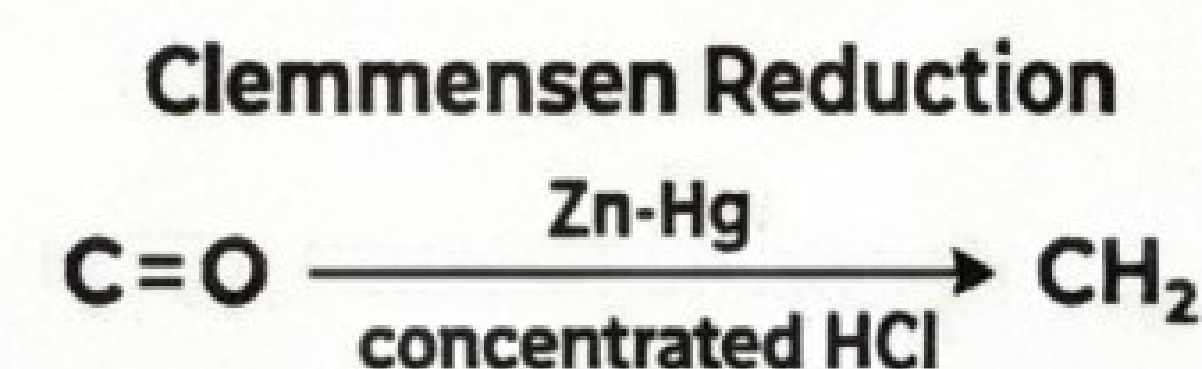
Occurs in aldehydes/ketones having at least one alpha-hydrogen. Reacts in dilute base.

Cannizzaro Reaction



Occurs in aldehydes with NO alpha-hydrogen. Reacts in concentrated base. One molecule oxidized to carboxylic salt, another reduced to alcohol.

Wolff-Kishner Reduction
using Hydrazine (NH₂-NH₂) and a strong base (KOH)



Clemmensen Reduction

ALL ABOUT CARBOXYLIC ACIDS

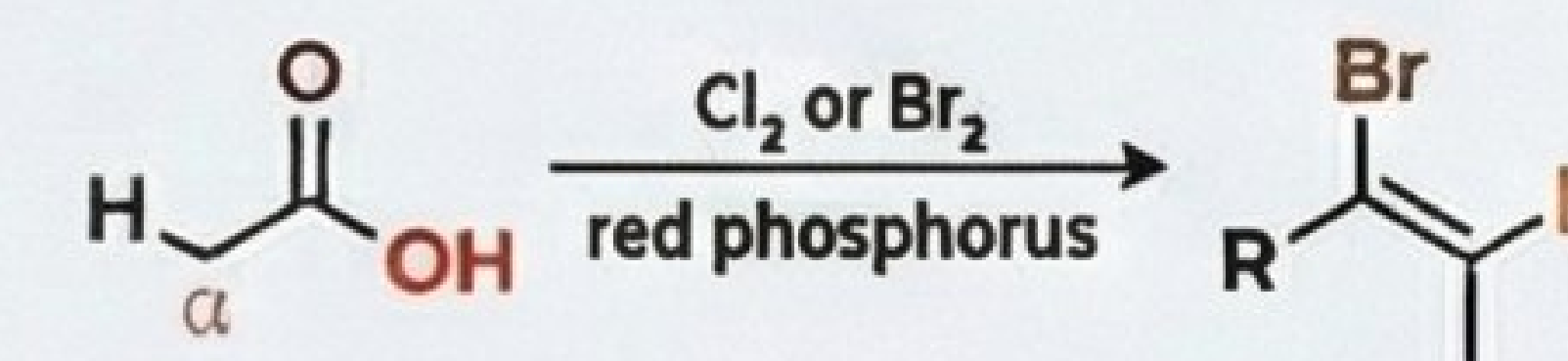
Comparing Acidic Strength

↑ Acidity increases with electron-withdrawing groups (-I effect, e.g., -NO₂, -F, -Cl) ↓ Acidity decreases with electron-donating groups (+I effect, e.g., -CH₃)

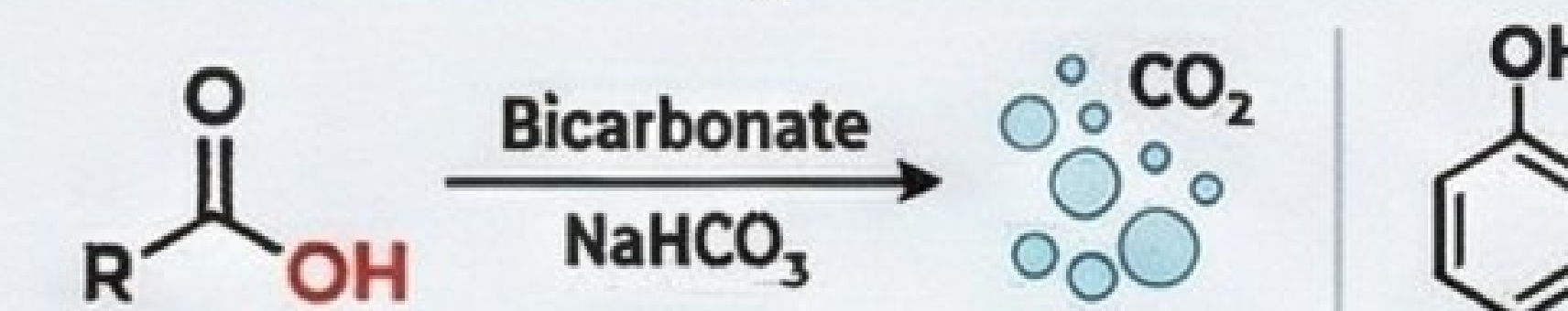
Relationship with pK_a

Carboxylic acids with an alpha-hydrogen react with Cl₂ or Br₂ in the presence of red phosphorus to substitute a halogen on the alpha-carbon.

Hell-Volhard-Zelinsky (HVZ) Reaction



Test to Distinguish from Phenol



Carboxylic acids react with Sodium Bicarbonate (NaHCO₃) to produce brisk effervescence (CO₂ gas). Phenols do not.